Gabriele Volani

Master Student in Computer Science | Software Developer @ Socialit

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Work Experience

Socialit - Full-stack Developer October 2024-Present

- Develop **full-stack web applications** taking **ownership** of end-to-end application evolution and using modern technologies and frameworks such as Spring and Angular.
- Collaborate with cross-functional teams and **interact directly with customers** to deliver software within project deadlines, supporting also the successful execution of **EU-based projects**.

Socialit - Software Developer Intern March 2024-May 2024

- Contributed to the development of a e-learning **CMS platform**, designing UI mockups, supporting database architecture and implementing authentication services and front-end pages.

Education

Expected 2027 **Master Degree in Computer Science - Data Science** **University of Trento**
Graduated 2024 **Bachelor Degree in Computer Science** **University of Trento**

Competitions and Programs

CyberChallenge.IT March 2026 - June 2026

- Selected for the CyberChallenge **cybersecurity training program**, completing an intensive hands-on program focused on ethical hacking and secure systems and participating in the **local competition**.

Public AI Challenge September 2023 - December 2023

- **Led** a team in developing a neural network to predict overtourism risks at critical Dolomite passes by integrating traffic, pedestrian and weather data.
- Managed the **end-to-end data science pipeline**, from data collection and preprocessing to feature engineering, model development and validation.

Selected Projects

High-Performance Parallel Numerical Integration

- Designed and benchmarked **scalable parallel algorithms** for 2D numerical integration on a 7,600+ core HPC cluster, leveraging MPI, OpenMP and hybrid programming techniques to optimize execution.

Blockchain-Based Academic Credential Verification System

- Developed EduVerify, a **blockchain-based academic credential verification** dApp on Polygon, implementing Solidity smart contracts with multi-signature governance, decentralized storage via IPFS and a React.js frontend integrated with Web3 technologies.

Fault-Tolerant Data Collection Protocol for Wireless Sensor Networks

- Developed and validated a **fault-tolerant IoT routing protocol** on Contiki OS for wireless sensor networks, implementing recovery mechanisms for node failures and achieving 99% PDR with up to 40× energy savings in real-world environments.

Skills

Languages	Java, Python, C, C++, Solidity, SQL, Typescript, Shell Scripting
Web programming	Spring Boot, JDBC, Hibernate, Express, NodeJS, Angular
Parallel programming and ML	PySpark, OpenMP, MPI, PyTorch
Infrastructure	Docker, Kubernetes, Git, Linux, Firebase, PBS Schedulers
Concepts	Microservices, Blockchain, Distributed Systems, IoT, HPC, Deep Learning